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High-frequency dynamics of three-dimensional beam trusses

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ABSTRACT: The theory of micro-local analysis of wave sytems shows that the energy density of three-dimensional Timoshenko beams associated with their high-frequency vibrations satisfies a Liouville-type transport equation. The material of the beam is assumed to be isotropic, and its heterogeneity varies slowly compared to the wavelength. Moreover the curvature and torsion of the beam are accounted for. The first part of the paper deals with the derivation of the transport model for a single three-dimensional beam. In order to extend this model to beam trusses, the reflection/transmission phenomena of the energy fluxes at junctions of beams are described by power flow reflection/transmission operators in a subsequent part. For numerical simulations, a discontinuous "Galerkin" finite element method is used on account of the discontinuities of the energy density field at the junctions. Thus a complete mechanical-numerical modelling of the linear transient dynamics of beam trusses is proposed. It is illustrated by a numerical example highlighting some remarkable features of high-frequency vibrations: the onset of a diffusive regime characterized by energy equi-partition rules at late times. Energy diffusion is prompted by the multiple reflection/transmission of waves at the junctions, with possible mode (polarization) conversions. This is the regime applicable to the Statistical Energy Analysis (SEA) of structural acoustics systems. The main purpose of this research is to develop an effective strategy to simulate and predict the transient response of beam trusses impacted by acoustic or mechanical shocks.

KEY WORDS: High-frequency; Transport model; Timoshenko beam.

1 INTRODUCTION

Spatial structures are often subjected to impulse loads which induce high-frequency (HF) wave propagation. Despite some recent researches, the characterization of the transient response to such loads remains a difficult problem. These loads can dammage the integrity of the structures or equipments attached to them. For example, the solar panels unfolding is set off by a pyrotechnic shock leading to propagation of HF waves in the panels. Another example is the pyrotechnic cut for the space launcher/satellite separation. Thus it is necessary to improve the prediction of the HF structural responses with efficient theoretical and numerical models.

In this research, we are interested in HF wave propagation within three-dimensional beam trusses. In this range, classical finite element methods are not applicable on account of the small wavelengths and high modal densities. Therefore several heuristic strategies have been developed by engineers. Global approaches like the Statistical Energy Analysis (SEA) [1] of structural-acoustic systems give information on the mean energy level in each substructures. Its main issue is the estimation of the some core parameters like the coupling loss factors or the injected powers. Local approaches like the Vibrational Conductivity Analogy (VCA) [2] have been also investigated. The point of issue concerning the existing local approaches is their extension to complex structures. Another typical problem with the modelization of HF wave propagation is the resolution of the eikonal equation which raises difficulties for the linear superposition of waves [3] and for the treatment of boundary conditions. This point is adressed using Gaussian beams as done by Bougacha *et al.* [4] for the boundary conditions in a convex domain; or using an alternative partial differential technique based on the Segment Projection Method [5] that is usable for non convex domain. This last method has been applied in the context of geometrical optics (for scalar waves), thus no polarisation and no conversion phenomena are considered. A modal approach was developed by Keller & Rubinov [6], and applied to coupled Timoshenko beams with several type of joints by Chen & Zhou [7] and Coleman & Schaffer [8]. This method gives an asymptotic estimate of the eigenvalues for multi-dimensional scattering problems. The results are valid even in a relatively low frequency range but give no information on the transient response. An other method to avoid the wavelength issue is to adopt an energy transport formulation. It is the method developed here.

Based on microlocal analysis of classical wave systems, it can be proved that the energy density associated with their HF solutions satisfies a Liouville-type transport equation. This theory can be adapted to quantum wave, classical acoustic or electromagnetic waves as shown by Papanicolaou & Ryzhik [3]. Transport equations can be solved numerically by various methods including Monte-Carlo methods [9] or nodal/spectral discontinuous Galerkin (DG) finite element methods [10]. Both techniques have the advantage to be weakly dissipative and weakly dispersive (depending on the order of interpolation). Thus they are well suited for late-time simulations in order to exhibit the diffusive behavior of the transport equations [11]. These methods have been applied to assemblies of two-dimensional beams or plates [12], as well as to a randomly heterogeneous shell [13]. The main objectives of this paper are twofold. First, the two-dimensional model of HF energy evolution derived by Savin [12] is extended to a curved, three-dimensional beam model for application to multi-bay trusses. Second, a general framework for numerical simulations by discontinous finite elements is derived. This mathematical-numerical model of the transient dynamics of beam trusses shall be able to predict, for example, their diffusive behavior at late times or the energy paths within the different bays. In the next section of this paper, the mechanical model of a curved, three-dimensional Timoshenko beam is outlined. The transport theory is introduced in section 3 and then applied to the Timoshenko beam model. In order to extend the theory to HF wave propagation in multibay trusses, the various reflection/transmission phenomena at junctions of beams have to be analyzed. This is the purpose of the third part of the paper. The numerical resolution of the transport equations by a discountinuous finite element method is considered in a subsequent section, where we illustrate this strategy with an application example. The last section offers a few conclusions.

2 THREE-DIMENSIONAL TIMOSHENKO BEAM MODEL

The application of the transport theory first requires to establish the dynamic behavior of a three-dimensional Timoshenko beam, i.e. its dynamic and constitutive equations. The purpose of this section is to summarize these equations prior to the HF setting proposed subsequently in section 3.

2.1 Geometry and kinematical hypotheses

The vibrational behavior of the beam is modeled by Timoshenko's beam kinematics. It includes the effects of rotary inertia and shear distortion [14]. It can be seen as a first order Taylor expansion in the radial coordinate of the motion with respect to the beam neutral fiber (defined below). Another important point in this theory is the introduction of a shear coefficient to account for the variation of the shear stress along the cross-section. The latter is assumed to be rigid and normal to the neutral fiber at rest. In this paper beam's material is assumed to be isotropic but not necessarily homogeneous. The heterogeneities vary slowly in regard to the considered wavelengths. The parameters describing the material are supposed to be symmetrically distributed on the cross-section. The geometrical and inertial centers of the cross-sections are assumed to be superimposed and form the neutral fiber. The model below allows for curvature and torsion of the latter. Let $\hat{\boldsymbol{t}}(s)$, $\hat{\boldsymbol{n}}(s)$, $\hat{\boldsymbol{b}}(s)$ be the tangent, normal and bi-normal unit vectors constituting a Frenet-Serret frame defined about the neutral fiber $\mathscr{C}$, which is parametrized by the curvilinear abscissa s in a (bounded or not) subset $\mathscr{S}$ of $\mathbb{R}$. The corresponding curvilinear coordinates are $\boldsymbol{s} = (s, s_2, s_3)^T$ such that a point $\boldsymbol{x}$ in the subset Ω of $\mathbb{R}^3$ occupied by the beam is parametrized by $\boldsymbol{x} = \Psi(\boldsymbol{s})$ in a fixed reference frame. Ω is also written $\Omega \equiv \mathscr{C} \times \Sigma(s)$, where Σ is the beam cross-section. Then $\hat{\boldsymbol{t}}(s) = d\boldsymbol{x}/ds|_{\boldsymbol{s}=(s,0,0)^T}$, $\hat{\boldsymbol{n}}(s) = |d\hat{\boldsymbol{t}}/ds|^{-1} d\hat{\boldsymbol{t}}/ds$, and $\hat{\boldsymbol{b}}(s) = \hat{\boldsymbol{t}} \times \hat{\boldsymbol{n}}$. The curvature $\kappa(s)$ is defined by $d\hat{\boldsymbol{t}}/ds = \kappa\hat{\boldsymbol{n}}$, and the torsion $\tau(s)$ is defined by $d\hat{\boldsymbol{b}}/ds = -\tau\hat{\boldsymbol{n}}$. The cross-section dimensions have to be small in regard to both the curvature and the torsion [15]. By Timoshenko's beam kinematical assumption, the displacement of a point $\boldsymbol{s}$ of the cross-section is written in the local frame:

$$\boldsymbol{u}(\boldsymbol{s},t) = \boldsymbol{u}_c(s,t) + \boldsymbol{\theta}(s,t) \times \boldsymbol{s}_\perp \,, \tag{1}$$

where $\boldsymbol{s}_\perp = \boldsymbol{s} - (\boldsymbol{s}\cdot\hat{\boldsymbol{t}})\hat{\boldsymbol{t}} \in \Sigma(s)$, the displacement of the neutral fiber is $\boldsymbol{u}_c(s,t) = u_c(s,t)\hat{\boldsymbol{t}}(s) + v_c(s,t)\hat{\boldsymbol{n}}(s) + w_c(s,t)\hat{\boldsymbol{b}}(s)$, and $\boldsymbol{\theta}(s,t) = \theta_1(s,t)\hat{\boldsymbol{t}}(s) + \theta_2(s,t)\hat{\boldsymbol{n}}(s) + \theta_3(s,t)\hat{\boldsymbol{b}}(s)$ is the small rotation vector of the cross-section. This model may be refined in order to be more accurate in the HF range. For example an additional degree of freedom associated with the warping of the cross-section can be introduced [16] or a third curvature associated with the twist of the beam [17]. These topic are subject of ongoing research.

2.2 Mechanical equations

The derivation of the dynamical equilibrium and the constitutive equations of a three-dimensional curved beam with isotropic and elastic material under the assumption of small displacement leads to:

$$\boldsymbol{DF} = \rho S \ddot{\boldsymbol{U}}_c \,, \tag{2}$$

$$\boldsymbol{DM} + \hat{\boldsymbol{t}} \times \boldsymbol{F} = \boldsymbol{J}\ddot{\boldsymbol{\Theta}} \,, \tag{3}$$

$$\boldsymbol{F} = S\boldsymbol{C}_1(\boldsymbol{DU}_c + \hat{\boldsymbol{t}} \times \boldsymbol{\Theta}) \,, \tag{4}$$

$$\boldsymbol{M} = \boldsymbol{C}_2\boldsymbol{D\Theta} \,, \tag{5}$$

where $\rho(s)$ is the mean material volume density and S is the cross-section area, $\boldsymbol{F}(s,t) = (T_1(s,t), T_2(s,t), T_3(s,t))^T$ and $\boldsymbol{M}(s,t) = (M_1(s,t), M_2(s,t), M_3(s,t))^T$ are respectively the resultant forces and resultant moments acting on the cross-section in the local frame, $\boldsymbol{U}_c$ and Θ are the vectors in $\mathbb{R}^3$ constituted by the coordinates of $\boldsymbol{u}_c$ and $\boldsymbol{\theta}$ in the local Frenet frame, $\boldsymbol{J}(s) = \mathrm{diag}(I(s), J(s), K(s))$, $J = \rho\int_\Sigma s_3^2 d\Sigma$, $K = \rho\int_\Sigma s_2^2 d\Sigma$, $I = J + K$ are the inertias of the beam, $\boldsymbol{C}_1(s) = \mathrm{diag}(E, \mu G, \mu G)$ is the relaxation tensor, with $(\mu G)(s)$ the reduced shear modulus averaged over the cross-section, and $E(s)$ is the Young modulus averaged over the cross-section, $\boldsymbol{C}_2(s) = \mathrm{diag}(\mu GI, EJ, EK)$ is the average relaxation tensor for rotational motions, and $\boldsymbol{D}(s)$ is the so-called Frenet matrix induced by the the derivation with respect to the curvilinear absissa along the neutral fiber:

$$\boldsymbol{D}(s) = \begin{bmatrix} \frac{\partial}{\partial s} & -\kappa & 0 \\ \kappa & \frac{\partial}{\partial s} & -\tau \\ 0 & \tau & \frac{\partial}{\partial s} \end{bmatrix} . \tag{6}$$

Double dots stand for double derivation with respect to time. Details of the calculation can be found in [16, 18, 19].

2.3 Energetic observables

The dynamics of the beam reduced to its neutral fiber $s \in \mathscr{S}$ by integration over the cross-section $\Sigma(s)$ is described by equations (2 - 5). Introducing the state vector $\boldsymbol{X}(s,t) = (\dot{\boldsymbol{U}}_c^T, \dot{\boldsymbol{\Theta}}^T, \boldsymbol{F}^T, \boldsymbol{M}^T)^T$ of $\mathbb{R}^{12}$, they can be written as an 12×12 first-order hyperbolic system:

$$\begin{cases} \boldsymbol{A}(s)\partial_t \boldsymbol{X} = \boldsymbol{P}(s, \partial_s)\boldsymbol{X} \,, \\ \boldsymbol{X}(0,s) = \boldsymbol{X}_0(s) \,, \end{cases} \tag{7}$$

where $\boldsymbol{A}$ is a block diagonal matrix defined by $\boldsymbol{A}(s) = \mathrm{diag}(\rho S\boldsymbol{I}_3, \boldsymbol{J}, (S\boldsymbol{C}_1)^{-1}, \boldsymbol{C}_2^{-1})$, $\boldsymbol{I}_n$ is the $n \times n$ identity matrix, and

$\boldsymbol{P}(s,\partial_s)$ is the operator:

$$\boldsymbol{P}(s,\partial_s)=\begin{bmatrix}0&0&\boldsymbol{D}&0\\0&0&\boldsymbol{\Omega}&\boldsymbol{D}\\\boldsymbol{D}&\boldsymbol{\Omega}&0&0\\0&\boldsymbol{D}&0&0\end{bmatrix}\,,\quad \boldsymbol{\Omega}=\begin{bmatrix}0&0&0\\0&0&-1\\0&1&0\end{bmatrix}\tag{8}$$

which may also be written $\boldsymbol{P}(s,\partial_s)=\boldsymbol{D}^1\partial_s+\boldsymbol{D}^0(s)$; $\boldsymbol{D}^1$ and $\boldsymbol{D}^0$ are 12×12 matrices obtained straightforwardly from equation (8) above and equation (6). Note that $\boldsymbol{D}^1$ is symmetric and has elements equal to 0 or 1, and $\boldsymbol{D}^0$ is skew-symmetric. Here $\boldsymbol{X}_0(s)$ is a vector of initial conditions. Then the mechanical energy density $\mathscr{E}\in\mathbb{R}_+$ for the beam motion is:

$$\mathscr{E}(s,t)=\frac{1}{2}(\boldsymbol{X},\boldsymbol{X})_{\boldsymbol{A}}\,,\tag{9}$$

and the power flow density $\Pi\in\mathbb{R}$ within the beam is:

$$\Pi(s,t)=-\frac{1}{2}(\boldsymbol{X},\boldsymbol{X})_{\boldsymbol{D}^1}=-(\boldsymbol{F}\cdot\dot{\boldsymbol{U}}_c+\boldsymbol{M}\cdot\dot{\boldsymbol{\Theta}})\,,\tag{10}$$

where we have introduced the notation $(\boldsymbol{u},\boldsymbol{v})_{\boldsymbol{G}}:=\boldsymbol{u}^T\boldsymbol{G}\boldsymbol{v}$ for a square matrix $\boldsymbol{G}$. Finally, taking the scalar product of equation (7) with $\boldsymbol{X}$ leads to the energy conservation law, or continuity equation:

$$\partial_t\mathscr{E}+\cdot\partial_s\Pi=0\,.\tag{11}$$

3 HIGH-FREQUENCY WAVE PROPAGATION IN A THREE-DIMENSIONAL BEAM

In this section, we are interested in the computation of the solution of the system (7) in the HF regime. High-frequency vibrations of the beam may be generated by initial conditions oscillating at a length scale ε comparable to the wavelength. The HF regime thus corresponds to $\varepsilon\to 0$. These initial conditions may be chosen for example as plane waves in the form $\boldsymbol{X}^0_\varepsilon(s)=\boldsymbol{\chi}(s)e^{\mathrm{i}ks/\varepsilon}$ where $k\in\mathbb{R}$ and $\boldsymbol{\chi}(s)$ is a slowly varying vector. As easily seen, it is not possible to describe the limit of $\boldsymbol{X}^0_\varepsilon$ as $\varepsilon\to 0$. Since by the hyperbolicity of equation (7) its solution $\boldsymbol{X}_\varepsilon(s,t)$ has the same shape as the initial condition, it is neither possible to describe its HF limit. However the energy input by the initial condition will propagate in the beam and one should be able to track it by equation (11), provided its limit when $\varepsilon\to 0$ is known. Therefore energetic quantities associated to $\boldsymbol{X}_\varepsilon$ shall be considered in this very limit. The link between the solution $\boldsymbol{X}_\varepsilon$ of equation (7) and its energy as $\varepsilon\to 0$ is established using a so-called Wigner transform of $(\boldsymbol{X}_\varepsilon)$ and its limit, the Wigner measure, as $\varepsilon\to 0$. The theoretical approach developed in [3, 20–22] outlines how this limit is obtained, and how a continuity equation of the form (11) is derived for the Wigner measure in phase space. Indeed, the main novelty of the proposed analysis is that all energetic quantities shall be resolved in $T^*\mathscr{S}=\mathscr{S}\times\mathbb{R}_k$ (the phase space) in order to correctly built up their HF limits and evolution properties. The general framework is described in section 3.1 below, and its application to a curved three-dimensional Timoshenko beam is given in section 3.2.

3.1 *High-frequency energy density and transport properties*

In this section, the HF limit of a first-order hyperbolic system of the form (7) is considered in a general setting. First, the Wigner transform and its properties are introduced. This transform is used to estimate the energy density associated to the solutions of the system (7). Let $\boldsymbol{X}(s,t)$ be a temperate distribution defined on $\mathbb{R}_s\times\mathbb{R}_t$ with values in $\mathbb{C}^n$, its spatial Wigner transform is defined by:

$$\boldsymbol{W}_\varepsilon[\boldsymbol{X}](s,k,t)=\frac{1}{2\pi}\int_{\mathbb{R}}e^{\mathrm{i}k\cdot s'}\boldsymbol{X}\left(s-\varepsilon\frac{s'}{2},t\right)\boldsymbol{X}^*\left(s+\varepsilon\frac{s'}{2},t\right)ds'\,,\tag{12}$$

where $\boldsymbol{X}^*=\overline{\boldsymbol{X}}^T$ is the conjugate transpose of $\boldsymbol{X}$. The $n\times n$ matrix $\boldsymbol{W}_\varepsilon[\boldsymbol{X}]$ is Hermitian and not necessarily positive but it does become positive in the HF limit $\varepsilon\to 0$, provided that $\boldsymbol{X}$ lies in the set of square integrable functions with respect to s. Furthermore this (weak) limit always exists in this case. The Wigner transform has the property:

$$\int_{\mathbb{R}}\boldsymbol{W}_\varepsilon[\boldsymbol{X}](s,k,t)dk=\boldsymbol{X}(s,t)\boldsymbol{X}^*(s,t)\,,\tag{13}$$

hence its link with the energy if one takes the trace of the above equation. Now let us consider the $\mathbb{C}^{12}$ vector $\boldsymbol{X}_\varepsilon$ satisfying the rescaled first-order hyperbolic problem (7):

$$\left\{\begin{array}{l}\varepsilon\boldsymbol{A}(s)\partial_t\boldsymbol{X}_\varepsilon=\boldsymbol{P}_\varepsilon(s,\varepsilon\partial_s)\boldsymbol{X}_\varepsilon\,,\\ \boldsymbol{X}_\varepsilon(0,s)=\boldsymbol{X}^0_\varepsilon(s)\,,\end{array}\right.\tag{14}$$

where $\boldsymbol{A}(s)$ is independent of ε (the medium is varying on a scale much longer than the small wavelength ε); also $\boldsymbol{P}_\varepsilon(s,\varepsilon\partial_s)=\boldsymbol{D}^1(\varepsilon\partial_s)+\varepsilon\boldsymbol{D}^0(s)$ where $\boldsymbol{D}^1$ is symmetric, and $\boldsymbol{D}^0$ is skew-symmetric and independent of ε. The initial data $\boldsymbol{X}^0_\varepsilon$ vary on a scale lower or equal to ε, which is the case of the plane waves $\boldsymbol{X}^0_\varepsilon(s)\propto e^{\mathrm{i}ks/\varepsilon}$ considered above. More generally, these data have to satisfy the so-called ε-oscillatory strong condition stating that $(|\varepsilon\partial_s|\boldsymbol{X}^0_\varepsilon)^2$ remains at least locally integrable on $\mathscr{S}$ [21]; HF plane waves are only a particular case satisfying this condition. Then provided that the sequence $(\boldsymbol{X}_\varepsilon)$ remains bounded in the set of square integrable functions with respect to $s\in\mathscr{S}$, its Wigner transform $\boldsymbol{W}_\varepsilon[\boldsymbol{X}_\varepsilon]$ has an Hermitian weak limit (denoted by $\boldsymbol{W}$) as $\varepsilon\to 0$ which is also non negative, the so-called Wigner measure of the sequence $(\boldsymbol{X}_\varepsilon)$; see [21, 22]. The energy density of equation (9) is expressed in terms of $\boldsymbol{W}$ by:

$$\mathscr{E}(s,t)=\frac{1}{2}\int_{\mathbb{R}}\mathrm{Tr}(\boldsymbol{A}\boldsymbol{W})\,dk,\tag{15}$$

and the power flow density equation (10) is:

$$\Pi(s,t)=-\frac{1}{2}\int_{\mathbb{R}}\mathrm{Tr}(\boldsymbol{D}^1\boldsymbol{W})\,dk\tag{16}$$

as $\varepsilon\to 0$. $\mathrm{Tr}\,\boldsymbol{G}$ denotes the trace of a square matrix $\boldsymbol{G}$. Since the sequence $(\boldsymbol{X}_\varepsilon)$ satisfies the system (14), it can be shown in addition that its Wigner measure can be written as (see [3, 13, 20]):

$$\boldsymbol{W}(s,k,t)=\sum_{\alpha=1}^{M}\sum_{i,j=1}^{R_\alpha}w^{ij}_\alpha(s,k,t)\boldsymbol{v}_{\alpha_i}(s,k)\otimes\boldsymbol{v}_{\alpha_j}(s,k)\,,\tag{17}$$

where $(\boldsymbol{v}_{\alpha_i})_{1\leq i\leq R_\alpha}$ are the eigenvectors of the dispersion matrix $\boldsymbol{L}(s,k):=k\boldsymbol{A}^{-1}(s)\boldsymbol{D}^1$ associated with the eigenvalue λ_α of which order of multiplicity is R_α (these orders of multiplicity are assumed to be independent of $(s,k)\in T^*\mathscr{S}\backslash\{k=0\}$ and are such that $\sum_{\alpha=1}^{M} R_\alpha = 12$). The $R_\alpha \times R_\alpha$ coherence matrices $\boldsymbol{W}_\alpha$, defined as $[\boldsymbol{W}_\alpha]_{ij} = w_\alpha^{ij}$, satisfy the transport equations:

$$\partial_t \boldsymbol{W}_\alpha + \{\lambda_\alpha, \boldsymbol{W}_\alpha\} + [\boldsymbol{W}_\alpha, \boldsymbol{N}_\alpha] = \boldsymbol{0}, \tag{18}$$

where $\{g,h\} = \partial_k g \cdot \partial_s h - \partial_s g \cdot \partial_k h$ is the usual Poisson bracket, $[\boldsymbol{G},\boldsymbol{H}] = \boldsymbol{GH} + \boldsymbol{H}^*\boldsymbol{G}^*$, and $\boldsymbol{N}_\alpha$ is an $R_\alpha \times R_\alpha$ skew-symmetric matrix, the so-called coupling matrix, with elements:

$$\begin{aligned}[\boldsymbol{N}_\alpha(s)]_{ij} = {}& (\boldsymbol{v}_{\alpha_j}, \boldsymbol{v}_{\alpha_i})_{\boldsymbol{P}} \\ & - \partial_s \lambda_\alpha (\partial_k \boldsymbol{v}_{\alpha_i}, \boldsymbol{v}_{\alpha_j})_{\boldsymbol{A}} - \frac{1}{2}(\partial_s \partial_k \lambda_\alpha)\delta_{ij}.\end{aligned} \tag{19}$$

3.2 Application to a three-dimensional Timoshenko beam

Let $\boldsymbol{k} := k\hat{\boldsymbol{t}}$ be the wave vector within a beam oriented by its tangent unit vector $\hat{\boldsymbol{t}}$. As $\boldsymbol{D}^0$ is rescaled as a first-order term in ε, it does not influence the eigenvalues of the zeroth-order dispersion matrix $\boldsymbol{L}(s,k)$ of the system (7). Thus the curvatures have no influence on the HF vibrations of the beam. The eigenvalues of $\boldsymbol{L}$ are:

$$\begin{aligned}\lambda_{\mathrm{P}}^{\pm}(s,\boldsymbol{k}) &= \pm c_{\mathrm{P}}(s)|\boldsymbol{k}| = \pm\sqrt{\frac{E}{\rho}}|\boldsymbol{k}| \qquad \text{each of multiplicity } 3, \\ \lambda_{\mathrm{T}}^{\pm}(s,\boldsymbol{k}) &= \pm c_{\mathrm{T}}(s)|\boldsymbol{k}| = \pm\sqrt{\frac{\mu G}{\rho}}|\boldsymbol{k}| \qquad \text{each of multiplicity } 3,\end{aligned} \tag{20}$$

where c_{P} and c_{T} are the group velocities of longitudinal and transverse waves, respectively (since these waves are non dispersive in the HF range, the group velocities are also the phase velocities). Hence $R_\alpha^{\pm} = 3$ for $\alpha = \mathrm{P},\mathrm{T}$ and either the forward $(+)$ or the backward $(-)$ waves. The associated eigenvectors, orthonormal with respect to the inner product $(\boldsymbol{u},\boldsymbol{v})_{\boldsymbol{L}}$, are:

$$\begin{aligned}
\boldsymbol{v}_{\mathrm{P}_1}^{\pm} &= \left(\frac{\mathrm{sign}(k)}{\sqrt{2\rho S}}, 0,0,0,0,0, \pm\sqrt{\frac{ES}{2}}, 0,0,0,0,0\right)^T, \\
\boldsymbol{v}_{\mathrm{P}_2}^{\pm} &= \left(0,0,0,0, \frac{\mathrm{sign}(k)}{\sqrt{2\rho J}}, 0,0,0,0,0, \pm\sqrt{\frac{EJ}{2}}, 0\right)^T, \\
\boldsymbol{v}_{\mathrm{P}_3}^{\pm} &= \left(0,0,0,0,0, \frac{\mathrm{sign}(k)}{\sqrt{2\rho K}}, 0,0,0,0,0, \pm\sqrt{\frac{EK}{2}}\right)^T, \\
\boldsymbol{v}_{\mathrm{T}_1}^{\pm} &= \left(0, \frac{\mathrm{sign}(k)}{\sqrt{2\rho S}}, 0,0,0,0,0, \pm\sqrt{\frac{\mu GS}{2}}, 0,0,0,0\right)^T, \\
\boldsymbol{v}_{\mathrm{T}_2}^{\pm} &= \left(0,0, \frac{\mathrm{sign}(k)}{\sqrt{2\rho S}}, 0,0,0,0,0, \pm\sqrt{\frac{\mu GS}{2}}, 0,0,0\right)^T, \\
\boldsymbol{v}_{\mathrm{T}_3}^{\pm} &= \left(0,0,0, \frac{\mathrm{sign}(k)}{\sqrt{2\rho I}}, 0,0,0,0,0, \pm\sqrt{\frac{\mu GI}{2}}, 0,0\right)^T.
\end{aligned} \tag{21}$$

The eigenvectors associated with the eigenvalues P$\pm$ are the compressional (1) and bending (2 and 3) modes, and the eigenvectors associated with the eigenvalues T$\pm$ are the shear (1 and 2) and torsional (3) modes. As these eigenvectors are linearly independent, no coupling occurs between modes in the HF range away from the boundaries and interfaces (their influence is analyzed in the next section). The 3×3 coherence matrices $\boldsymbol{W}_\alpha^{\pm}$, $\alpha = \mathrm{P},\mathrm{T}$, satisfy the Liouville-type transport equations:

$$\partial_t \boldsymbol{W}_\alpha^{\pm} \pm \mathrm{sign}(k) c_\alpha \cdot \partial_s \boldsymbol{W}_\alpha^{\pm} \mp |\boldsymbol{k}| \partial_s c_\alpha \cdot \partial_k \boldsymbol{W}_\alpha^{\pm} = \boldsymbol{0}, \tag{22}$$

as one can check that the brackets $[\boldsymbol{W}_\alpha^{\pm}, \boldsymbol{N}_\alpha^{\pm}]$ vanish for beams. These results extend to three dimensions the derivation of Savin [12] for two-dimensional beam trusses. Accordingly, it can be observed that $\boldsymbol{W}_\alpha^{-}(s,k,t) = \boldsymbol{W}_\alpha^{+}(s,-k,t)$ for $\alpha = \mathrm{P},\mathrm{T}$, so that the space time energy density reduces to:

$$\mathscr{E}(s,t) = \sum_{\alpha=\mathrm{P},\mathrm{T}} \int_{\mathbb{R}} \mathrm{Tr}\, \boldsymbol{W}_\alpha^{+}(s,k,t)\, dk, \tag{23}$$

and the power flow density vector $\boldsymbol{\Pi} = \Pi\hat{\boldsymbol{t}}$ is:

$$\begin{aligned}\boldsymbol{\Pi}(s,t) &= \sum_{\alpha=\mathrm{P},\mathrm{T}} c_\alpha(s) \int_{\mathbb{R}} \mathrm{Tr}\, \boldsymbol{W}_\alpha^{+}(s,k,t)\hat{\boldsymbol{k}}\, dk \\ &= \sum_{\alpha=\mathrm{P},\mathrm{T}} \sum_{j=1}^{R_\alpha} \int_{\mathbb{R}} \boldsymbol{\pi}_\alpha^{jj}(s,\boldsymbol{k},t)\, dk,\end{aligned} \tag{24}$$

where $\boldsymbol{\pi}_\alpha^{jj}(s,\boldsymbol{k},t) := c_\alpha(s) w_\alpha^{jj}(s,k,t)\hat{\boldsymbol{k}}$ is the power flow density of a mode $\alpha \in \{\mathrm{P},\mathrm{T}\}$, $1 \leq j \leq R_\alpha$, in a beam oriented by its tangent unit vector $\hat{\boldsymbol{t}}$, with $\hat{\boldsymbol{k}} := \frac{\boldsymbol{k}}{|\boldsymbol{k}|} = \mathrm{sign}(k)\hat{\boldsymbol{t}}$: if $k > 0$ the energy flux travels in the same direction as $\hat{\boldsymbol{t}}$, but if $k < 0$ it travels in the opposite direction. For convenience the sign function is denoted by $\mathrm{sign}(k) = \hat{k}$ in the remaining of the paper. The energy and power flow densities (equation (23) and equation (24)) one ultimately wants to compute depend on the diagonal elements w_α^{jj} of $\boldsymbol{W}_\alpha$ solely. The transport equations (22) do not couple them (but the boundary/interface conditions considered in the next section do), so we only have to consider equations (22) for the elements $(w_\alpha^{jj})^+$ of $\boldsymbol{W}_\alpha^{+}$, reminding the symmetry $\boldsymbol{W}_\alpha^{+}(s,k,t) = \boldsymbol{W}_\alpha^{-}(s,-k,t)$. Introducing the set of energy modes $E = \{\mathrm{P}_1,\mathrm{P}_2,\mathrm{P}_3,\mathrm{T}_1,\mathrm{T}_2,\mathrm{T}_3\}$ corresponding to the eigenvectors $\boldsymbol{v}_{\mathrm{P}_j}^{+}$ and $\boldsymbol{v}_{\mathrm{T}_j}^{+}$, $1 \leq j \leq 3$, and the eigenvalues λ_{P}^{+} and λ_{T}^{+} (counted with their orders of multiplicity), the set of transport equations to be considered is finally reduced to:

$$\partial_t w_\alpha + c_\alpha \hat{k}\, \partial_s w_\alpha - |\boldsymbol{k}| \partial_s c_\alpha\, \partial_k w_\alpha = 0, \quad \alpha \in E. \tag{25}$$

Accordingly, the power flow density of a mode $\alpha \in E$ is denoted by:

$$\boldsymbol{\pi}_\alpha(s,\boldsymbol{k},t) = c_\alpha(s) w_\alpha(s,k,t)\hat{\boldsymbol{k}}. \tag{26}$$

4 HIGH-FREQUENCY POWER FLOW COEFFICIENTS AT THE JUNCTIONS

The HF energy density propagation in a beam is described by the transport equations (25). In order to use this model for complex three-dimensional beam trusses, the phenomena at the junctions have to be considered. They contribute to couple the energy densities w_α. This section deals with the energy density propagation in a junction of an arbitrary number $\mathscr{N}$ of beams. Power flow reflection/transmission coefficients are derived along the same lines as in Savin [12] for two-dimensional junctions.

4.1 Energy transport in coupled structures

Let us consider $\mathscr{N}$ beams occupying the domains Ω_p of $\mathbb{R}^3$, $1 \leq p \leq \mathscr{N}$. They are parametrized by their curvilinear coordinates $\boldsymbol{s} = (s, s_2, s_3)^T \in \mathbb{R}^3$ such that $s \in \mathscr{S}_p \subset \mathbb{R}$ for beam #p, where $\mathscr{S}_p$, $1 \leq p \leq \mathscr{N}$, is an interval of $\mathbb{R}$ such that $\mathscr{C}_p = \Psi_p(\mathscr{S}_p)$ is the curved line in $\mathbb{R}^3$ constituting the neutral fiber of beam #p. Their junction $\Gamma := \cap_{p=1}^{\mathscr{N}} \mathscr{C}_p$ reduces to a single point $\boldsymbol{x}_0 = \Psi_p(\boldsymbol{s}_0^p)$ in this parametrization. Let $\hat{\boldsymbol{t}}^p$ be the tangent vector to $\mathscr{C}_p$, $1 \leq p \leq \mathscr{N}$, pointing outward from beam #p at the junction, then it is assumed for convenience that $\boldsymbol{s}_0^p$ may be written $\boldsymbol{s}_0^p = s_0 \hat{\boldsymbol{t}}^p$ in the local Frenet frame of each beam #p. Owing to the Rankine-Hugoniot condition written for the transport equations (25), the power flow (24) is conserved across the junction [23]. For example, the energy density traveling in beam #1 and impinging the junction at $\boldsymbol{s}_0$ with an incident wave vector $\boldsymbol{k}$ such that $\boldsymbol{k} \cdot \hat{\boldsymbol{t}}^1 > 0$ is partially reflected and partially transmitted while the overall power flow remains constant. Then the flux flowing away from that junction in beam #p after reflection and transmission is written:

$$|\boldsymbol{\pi}_\alpha^p(s_0, \boldsymbol{k}, t) \cdot \hat{\boldsymbol{t}}^p| = \sum_{\beta \in E} \Big(\rho_{\alpha\beta}^{pp}(s_0) \boldsymbol{\pi}_\beta^p(s_0, \boldsymbol{k}_\beta^p, t) \cdot \hat{\boldsymbol{t}}^p + \sum_{\substack{q=1 \\ q \neq p}}^{\mathscr{N}} \tau_{\alpha\beta}^{pq}(s_0) \boldsymbol{\pi}_\beta^q(s_0, \boldsymbol{k}_\beta^q, t) \cdot \hat{\boldsymbol{t}}^q \Big) \quad (27)$$

for $\hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{t}}^p < 0$, where $\boldsymbol{k}_\beta^q = |\boldsymbol{k}_\beta^q| \hat{\boldsymbol{t}}^q$ such that $c_\alpha^p(s_0)|\boldsymbol{k}_\alpha^p| = c_\beta^q(s_0)|\boldsymbol{k}_\beta^q| := \omega$ for $1 \leq p, q \leq \mathscr{N}$, $\alpha, \beta \in E$ by the conservation of the power flow through the junction, and the left traces of $\boldsymbol{\pi}_\alpha$ in beam #p at the junction are denoted by:

$$\boldsymbol{\pi}_\alpha^p(s_0, \boldsymbol{k}, t) := \lim_{h \downarrow 0} \boldsymbol{\pi}_\alpha(\boldsymbol{s}_0 \cdot \hat{\boldsymbol{t}}^p - h, \boldsymbol{k}, t) . \quad (28)$$

$\rho_{\alpha\beta}^{pp}(s_0)$ and $\tau_{\alpha\beta}^{pq}(s_0)$ are power flow reflection/transmission coefficients of the junction respectively, also called efficiencies in the dedicated literature. The above relations constitute the boundary/interface conditions to be used to solve the transport equations (25). It should be noted that they remain valid everywhere within the beams if the convention $\rho_{\alpha\beta}^{pp} = 0$ and $\tau_{\alpha\beta}^{pp} = \delta_{\alpha\beta}$ is adopted for all $s \neq s_0$.

4.2 Computation of reflection/transmission coefficients for a beam junction

This section outlines how the power flow reflection/transmission coefficients $\rho_{\alpha_i\beta_j}^{pp}(s_0)$ and $\tau_{\alpha_i\beta_j}^{pq}(s_0)$ may be computed. A junction of $\mathscr{N}$ beams is considered. It is geometrically described by the Euler angles between beam #p and beam #q denoted by ψ_{pq}, θ_{pq} and φ_{pq} with the usual convention $(\hat{\boldsymbol{n}}, \hat{\boldsymbol{b}}, \hat{\boldsymbol{t}})$ meaning that the composition of three intrinsic rotations about the moving frame axes is used to rotate the local Frenet frame of beam #p to the local frame of beam #q; this rotation is denoted by $\boldsymbol{R}^{qp}$. The junction is at $s = s_0$, and it is assumed without loss of generality in the remaining that $s_0 = 0$. It is also considered that waves are incident in beam #1. Figure 1 shows an example of a junction of three beams, and the existing waves in each beam. From the subsection (3.2), we conclude that the HF wavenumber for compressional and bending motions is $k_{\mathrm{P}}^p = \omega / c_{\mathrm{P}}^p$, where ω is the circular frequency, and the HF

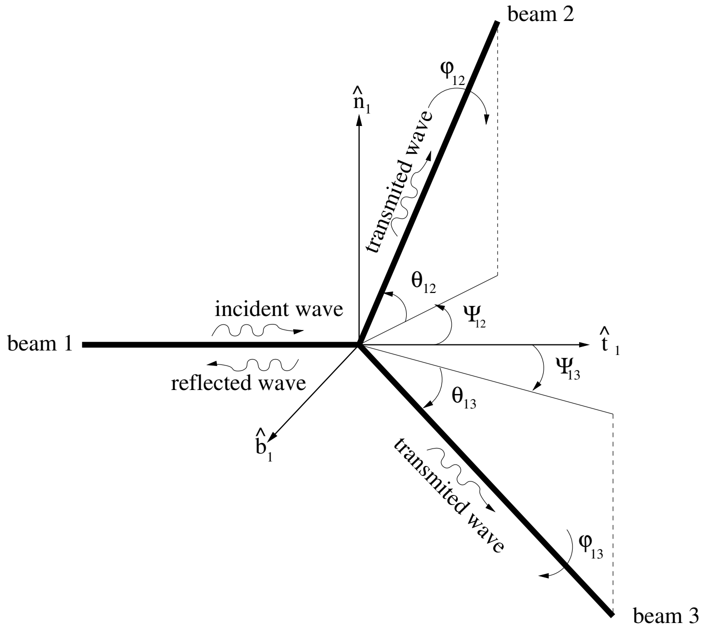

Figure 1. A junction of three coupled beams.

wavenumber for torsional and shear motions is $k_{\mathrm{T}}^p = \omega / c_{\mathrm{T}}^p$. The incident waves, traveling in beam #1 in the direction of increasing s may be written as $\exp^{-\mathrm{i}k_\alpha^1 s}$ for the translational waves ($\alpha = \mathrm{P}_1, \mathrm{T}_1, \mathrm{T}_2$) and $-\mathrm{i}k_\alpha^1 \exp^{-\mathrm{i}k_\alpha^1 s}$ for the rotational waves ($\mathrm{P}_2, \mathrm{P}_3, \mathrm{T}_3$). The corresponding reflected waves traveling in beam #1 in direction of decreasing s are written $B_\alpha \exp^{\mathrm{i}k_\alpha^1 s}$ for the translational waves and $B_\alpha \exp^{\mathrm{i}k_\alpha^1 s}$ for the rotational waves. The transmitted waves traveling in beam #q in direction of increasing s are written $C_\alpha^q \exp^{-\mathrm{i}k_\alpha^q s}$ for the translational waves and $-\mathrm{i}k_\alpha^q C_\alpha^q \exp^{-\mathrm{i}k_\alpha^q s}$ for the rotational waves. Coefficients B_i and C_i^q are obtained from the continuity of displacements, rotations, efforts and moments at the junction:

$$\begin{aligned}
\boldsymbol{U}_c^I(0) + \boldsymbol{U}_c^R(0) &= \boldsymbol{R}^{1q} \boldsymbol{U}_c^{Tq}(0) , \\
\boldsymbol{\Theta}^I(0) + \boldsymbol{\Theta}^R(0) &= \boldsymbol{R}^{1q} \boldsymbol{\Theta}^{Tq}(0) , \\
\boldsymbol{F}^1(0) &= \sum_{q=2}^{\mathscr{N}} \boldsymbol{R}^{1q} \boldsymbol{F}^q(0) , \\
\boldsymbol{M}^1(0) &= \sum_{q=2}^{\mathscr{N}} \boldsymbol{R}^{1q} \boldsymbol{M}^q(0) .
\end{aligned} \quad (29)$$

where $\boldsymbol{F}^q$ and $\boldsymbol{M}^q$ corresponds to the net force and net moment in beam #q, respectively, and $\boldsymbol{R}^{1q}$ is the rotation matrix corresponding to the projection of the frame of beam #q on the frame of beam #1. Inserting the constitutive equations (2 - 5) into equation (29) yields a linear system in terms of displacements and rotations solely. It can be easily solved for each type of incident waves and allows to compute the coefficients B_i, C_i^q. To be consistent with the transport model (22) governing the phase space energy densities, the associated power flows are derived from the generic formula for the energy flux of a HF plane wave propagating in a beam [12]. The time average power flow associated to a plane wave is:

$$< \Pi^q >= \frac{1}{2} \mathfrak{Re}\{\mathrm{i}\omega(\boldsymbol{F}^q \cdot \overline{\boldsymbol{U}_c^q} + \boldsymbol{M}^q \cdot \overline{\boldsymbol{\Theta}^q})\} ,$$

where $\mathfrak{Re}\{\boldsymbol{z}\}$ denotes the real part of $\boldsymbol{z}$. The HF power reflection/transmission coefficients for coupled Timoshenko beams are defined by:

$$\rho_{\alpha\beta}^{11} = \frac{< \Pi_\alpha^R >}{< \Pi_\beta^I >}, \tag{30}$$

$$\tau_{\alpha\beta}^{q1} = \frac{< \Pi_\alpha^{Tq} >}{< \Pi_\beta^I >}, \tag{31}$$

respectively, where $< \Pi_\alpha^R >$ is the time average power flow for the reflected waves corresponding to the mode $\alpha \in E$, $< \Pi_\alpha^{Tq} >$ is the time average power flow for the transmitted waves corresponding to the mode α in beam #q, and $< \Pi_\beta^I >$ is the time average power flow for an incident wave in the mode β. It is observed by computation that, in the HF limit, the reflection/transmission coefficients corresponding to translation incident motions u_c^I, v_c^I, or w_c^I, and rotational reflected/transmitted motions θ_i^R, θ_i^{Tq}, $i = 1,2$ or 3, are null; and those related to rotational incident motions and translational reflected/transmitted motions vanish as well. All other coefficients are independent of the circular frequency ω. Moreover, the sum of the power flow reflection/transmission coefficients is equal to one for a given angle.

5 NUMERICAL SIMULATION

Discountinities in the velocity field at the junction do not allow to use classical finite element method to solve numerically the Liouville equations (25) with the interface condition of equation (27). Indeed, although the overall power flow density is continuous across the junction owing to the Rankine-Hugoniot condition, the energy density itself does not necessarily have to be continuous across that junction. Therefore a discountinuous "Galerkin" (DG) finite element method [10, 24, 25] is used to solve equation (25) together with the reflection/transmission conditions of equation (27). The theoretical background of the DG method is recalled in section 5.1 below, while some numerical implementation issues are adressed in section 5.2. An illustrative example is presented in a subsequent section 5.3.

5.1 Discountinuous "Galerkin" finite elements

The DG method for the transport equations (25) is first written on a single beam $\mathscr{C}$ with $s \in \mathscr{S}$. Let us consider the subdivision $\mathscr{T}_h = \cup_{r=1}^{\mathsf{K}} \mathsf{D}_r$ of $\mathscr{S}$ (which is now assumed to be bounded) into K non-overlapping sub-domains, or finite elements D_r. These elements are written $\mathsf{D}_r = [s_{r-\frac{1}{2}}, s_{r+\frac{1}{2}}]$ and the largest element size of this mesh is $h = \max_{1\leq r\leq \mathsf{K}} |s_{r+\frac{1}{2}} - s_{r-\frac{1}{2}}|$. It is assumed that the velocities c_α are constant over the elements; they are denoted by c_α^r. Here $|k|$ is a fixed parameter, and the w_α's are considered as functions of $\hat{k}$ rather than k. A Galerkin variational formulation of the equations (25) in an *ad hoc* functional set V_h defined on $\mathscr{T}_h$ is: Find $w_\alpha \in \mathsf{V}_h$, $\alpha \in E$, such that

$$\int_{\mathsf{D}_r} \left(\frac{\partial w_\alpha}{\partial t}(s,\hat{k},t) v(s) - c_\alpha^r \hat{k} w_\alpha(s,\hat{k},t) \frac{dv}{ds} \right) ds = -\left[c_\alpha^r \hat{k} w_\alpha v \right]_{s_{r-\frac{1}{2}}}^{s_{r+\frac{1}{2}}} = -\left[\boldsymbol{\pi}_\alpha \cdot \hat{\boldsymbol{t}}^r v \right]_{s_{r-\frac{1}{2}}}^{s_{r+\frac{1}{2}}}, \quad \forall v \in \mathsf{V}_h, \tag{32}$$

recalling the definition of equation (26) for the power flow densities $\boldsymbol{\pi}_\alpha(s,\hat{\boldsymbol{k}},t)$, $\hat{\boldsymbol{t}}^r$ being the outward unit normal to $\partial\mathsf{D}_r = \{s_{r-\frac{1}{2}}, s_{r+\frac{1}{2}}\}$. As the sought solutions w_α of the above formulation may be discontinuous at a junction of beams, or at the interfaces between elements, the globally defined space V_h is introduced as $\mathsf{V}_h = \bigoplus_{r=1}^{\mathsf{K}} \mathsf{V}_h^r$ where the locally defined test spaces $\mathsf{V}_h^r \subseteq H^1(\mathsf{D}_r)$ do not enforce any particular boundary conditions on $\partial\mathsf{D}_r$. As a consequence of the lack of such conditions for the local solution and the test functions, the former is *a priori* multiply defined at the interfaces between elements (possibly a junction) and the boundary flux $\boldsymbol{\pi}_\alpha \cdot \hat{\boldsymbol{t}}^r$ in equation (32) is not defined. That is why it is replaced by a numerical flux $\boldsymbol{\pi}_\alpha^*$ which depends on the traces of w_α from both sides of an interface between elements. This leads to:

$$\int_{\mathsf{D}_r} \left(\frac{\partial w_\alpha}{\partial t}(s,\hat{k},t) v(s) - c_\alpha^r \hat{k} w_\alpha(s,\hat{k},t) \frac{dv}{ds} \right) ds = -\left[\boldsymbol{\pi}_\alpha^* \cdot \hat{\boldsymbol{t}}^r v \right]_{s_{r-\frac{1}{2}}}^{s_{r+\frac{1}{2}}}, \forall v \in \mathsf{V}_h. \tag{33}$$

The numerical flux $\boldsymbol{\pi}_\alpha^*$ at the interface $s = s_{r+\frac{1}{2}}$, say, is constructed from the computed values of $w_\alpha(s_{r+\frac{1}{2}},\hat{k},t)$ in V_h^r (the trace of w_α at $s_{r+\frac{1}{2}}$ in D_r) and $w_\alpha(s_{r+1-\frac{1}{2}},\hat{k},t)$ in V_h^{r+1} (the trace of w_α at $s_{r+\frac{1}{2}}$ in D_{r+1}) so as to be consistent and conservative, see [18] for more details.

5.2 Numerical implementation

Three main ingredients are needed to complete the numerical implementation of the DG method once the subdivision $\mathscr{T}_h$ has been performed: the definition of the numerical flux function $\boldsymbol{\pi}^*$ on all elements, the choice of the approximation set V_h, and the choice of a time-integration scheme for equation (33). The numerical fluxes ensuring the stability and consistency of the discountinuous Galerkin method for transport equations have the form [24, 25]:

$$\boldsymbol{\pi}^*(s_{r+\frac{1}{2}}; v^r, v^{r+1}) = \frac{1}{2}(v^r + v^{r+1})\hat{\boldsymbol{k}} + A_r(v^r - v^{r+1})\hat{\boldsymbol{t}}^r,$$

where A_r is a positive scalar. For example, the upwind flux corresponds to $A_r = \frac{1}{2}$. Invoking equation (27), this expression is generalized as follows for an interface (a junction) $\{s_{r+\frac{1}{2}}\} = \{s_{r'-\frac{1}{2}}\} = \partial\mathsf{D}_r \cap \partial\mathsf{D}_{r'}$, $r \neq r' \in \mathscr{I}_r$, where the group velocities c_α are discontinuous:

$$\boldsymbol{\pi}_\alpha^*(s_{r+\frac{1}{2}},\boldsymbol{k},t)\cdot\hat{\boldsymbol{t}}^r = \sum_{\beta\in E} \left(\rho_{\alpha\beta}^{rr} \boldsymbol{\pi}_\beta^r(s_{r+\frac{1}{2}},\boldsymbol{k}_\beta^r,t)\cdot\hat{\boldsymbol{t}}^r + \sum_{r'\in\mathscr{I}_r} \tau_{\alpha\beta}^{rr'} \boldsymbol{\pi}_\beta^{r'}(s_{r'-\frac{1}{2}},\boldsymbol{k}_\beta^{r'},t)\cdot\hat{\boldsymbol{t}}^{r'} \right), \tag{34}$$

for $\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{t}}^r < 0$, and:

$$\boldsymbol{\pi}_\alpha^*(s_{r+\frac{1}{2}},t)\cdot\hat{\boldsymbol{t}}^r = \boldsymbol{\pi}_\alpha^r(s,\boldsymbol{k},t)\cdot\hat{\boldsymbol{t}}^r, \tag{35}$$

for $\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{t}}^r > 0$. $\mathscr{I}_r$ is the set of indices of the beam elements which are connected to the beam element #r. This choice corresponds to an upwind numerical flux whenever $\rho_{\alpha\beta}^{rr} \equiv 0$ and $\tau_{\alpha\beta}^{rr'} \equiv \delta_{\alpha\beta}$ at any point s which is not a beam junction, in which case $\mathscr{I}_r = \{r+1\}$. The approximation set V_h is constructed as follows:

$$\mathsf{V}_h = \{v|_{\mathsf{D}_r} \in \mathbb{P}_\ell(\mathsf{D}_r), 1 \leq r \leq \mathsf{K}\}, \tag{36}$$

where $\mathbb{P}_\ell(\mathsf{D}_r) \subset H^1(\mathsf{D}_r)$ is typically a polynomial set. In our numerical simulations, $\mathbb{P}_\ell(\mathsf{D}_r)$ is spanned by a set of normalized Jacobi polynomials up to the order ℓ. The numerical dispersion and dissipation errors introduced by this spatial discretization are $\mathrm{O}(K^{2\ell+3})$ for decentered numerical fluxes ($A_r \neq 0$), or $\mathrm{O}(K^{4\lfloor \frac{\ell}{2} \rfloor+3})$ for centered fluxes ($A_r = 0$), [26] where $K = h|k|$ and $\lfloor \cdot \rfloor$ stands for the integer part. These so-called "superconvergence" properties are very much desirable for long time simulations in order to exhibit the diffusion limit of the transport equations (25) with the boundary conditions (27). Finally, time integration of the semi-discretized system (33) is performed by a strong stability-preserving high-order Runge-Kutta (RK) scheme [27].

5.3 *Numerical example*

A beam truss constituted by $\mathscr{N} = 13$ three-dimensional beams is considered: its shape is a cube having a side length $L = 5$ unit length ($L_p = L$ for $1 \leq p \leq 12$) and the 13-th beam is a transverse one linking two opposite vertices of this cube through its diagonal: thus its length is $L_{13} = \sqrt{3}L$ (see figure 2). All beams have the same cross-sections of area S and are made from the same homogeneous, isotropic elastic material. Thus all beams have the same velocities c_P and c_T for the longitudinal-bending and transverse-torsional modes, respectively. The Poisson ratio is $\nu = 0.3$ and the shear reduction factor is $\mu = 5(1+\nu)/6+5\nu$. The transport equations (25) hold in each beam and the reflection/transmission coefficients are directly computed from the approach outlined in section 4.2. The forward and backward flows are written as the superposition of reflected and transmitted flows including all existing energy modes, $\alpha = \mathrm{P}_j, \mathrm{T}_j$, $j = 1,2,3$, as described in section 4.1. The energetic initial condition has a Gaussian shape and loads the mode P_1 (a pure compressional wave). It is applied to, say, beam labelled #1 with a fixed initial direction $\hat{k} = +1$. The nodal density is fixed at $20/L$ uniform spatial elements per unit length for all beams. Legendre polynomials up to the third order ($\ell = 3$) are used as local basis functions, and the integration in time is performed by a fourth-order Runge-Kutta-Fehlberg scheme. The time scale $T = L/c_\mathrm{T}$ is introduced: it is the time needed by a transverse wave to travel across the side of the cube. This parameter has been set to $T = 50$ in our simulations. The Courant number $\mathrm{CFL} := c_\mathrm{T}\Delta t/h$ has been fixed to $\mathrm{CFL} = 0.16$. Figure 3 shows the evolution of the total energy $\int_0^{L_p} \mathscr{E}(s,t)ds$ for each beam, $1 \leq p \leq 13$. It is first noticed that the energetic behavior of parallel beams is similar at late times; hence, for the sake of clarity, the results of figure 3 are displayed for each beam group, and then the sums of the energies for each group. The first group is constituted by the beams parallel to the initially loaded beam #1 (beams #1, 5, 10, 11), the second group is constituted by the vertical beams #2, 4, 6, 8, and the third group is constituted by the remaining beams #3, 7, 9, 12 whereas the diagonal beam #13 is considered apart. At late times, the second and the third beam group have the same behavior for the total energy, because they are both perpendicular to the beam #1. It should be observed that the total energy for the entire truss is virtually conserved; this is a required property of the numerical scheme retained (the theoretical numerical dissipation error being about 4.10^{-9}). Moreover, the total energy in each beam tends to a limit as $t \to \infty$ which apparently depends on its mechanical parameters. This phenomenom exhibits the diffusive regime holding at late times. Finally, figure 4 shows the ratio between the overall transverse and longitudinal energy per beam $\mathscr{E}_\mathrm{T}^p(t)/\mathscr{E}_\mathrm{P}^p(t)$, where:

$$\mathscr{E}_\alpha^p(t) = \sum_{j=1}^{3} \sum_{\hat{k}=\pm 1} \int_0^{L_p} w_{\alpha_j}(s,\hat{k},t)ds \quad 1 \leq p \leq 13\,, \quad \alpha = \mathrm{P},\mathrm{T}\,.$$

This ratio converges to a constant value of the form $n_\mathrm{T}^p c_\mathrm{P}/n_\mathrm{P}^p c_\mathrm{T}$, where n_P^p and n_T^p are the numbers of compressional and transverse modes generated in each beam by the reflection/transmission processes at the junctions. n_P^p and n_T^p depend on the initial condition and the shape of the truss. However this limit should be analyzed theoretically more in detail; this issue is out of the scope of this paper but it is the subject of ongoing researches. For example, the initial motion in figure 3 and figure 4 is a compressional mode $\alpha = \mathrm{P}_1$ in beam #1; it is transmitted as transverse shear modes ($\alpha = \mathrm{T}_1$ or T_2) in the beams perpendicular to it (beams #2 and 12), and as transverse shear modes and the same compressional mode in beam #13; however no conversion to bending ($\alpha = \mathrm{P}_2$ or P_3) or torsional ($\alpha = \mathrm{T}_3$) modes occurs at any time. Thus $n_\mathrm{P} = 1$ and $n_\mathrm{T} = 2$ in this case. The strong oscillations at the begining of the simulation occur because some beams are not loaded or loaded with compressional or transverse waves exclusively.

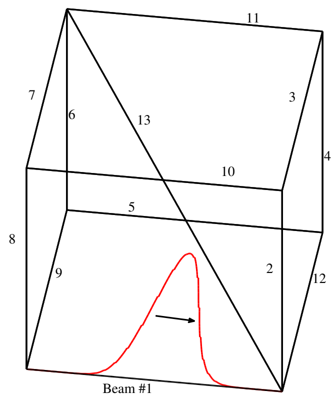

Figure 2. A view of the geometry of the beam truss and the initial condition in beam #1.

6 CONCLUSION

A transient transport model for the propagation of the HF vibrational energy density in three-dimensional beam trusses has been derived. It accounts for the power flow reflection and transmission phenomena at the junctions in this range. The proposed analytical model is integrable by a Runge-Kutta discountinuous Galerkin (RKDG) finite element scheme. Owing to its low numerical dispersion and dissipation errors, the RKDG method is suitable for long time simulations. Numerical results obtained for an example of a beam truss illustrate the transient regime and the onset of a diffusive regime at late times. The latter is prompted by the multiple reflections and transmissions of the energy modes at the beam junctions, with possible mode conversions. This scattering process is different

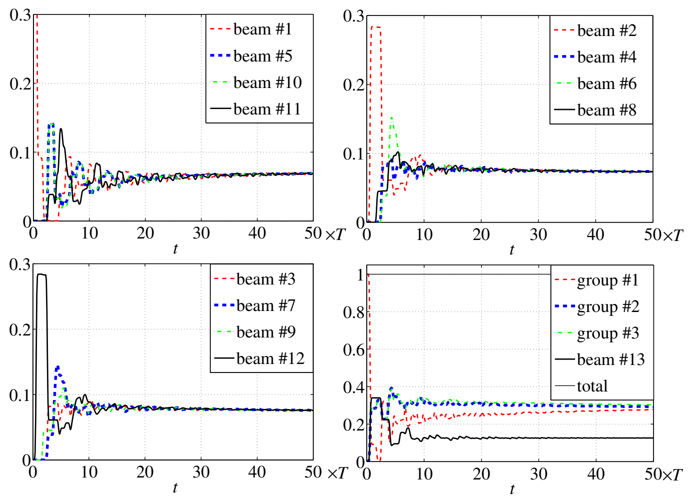

Figure 3. Evolution of the total vibrational energy in the truss for an incident longitudinal motion.

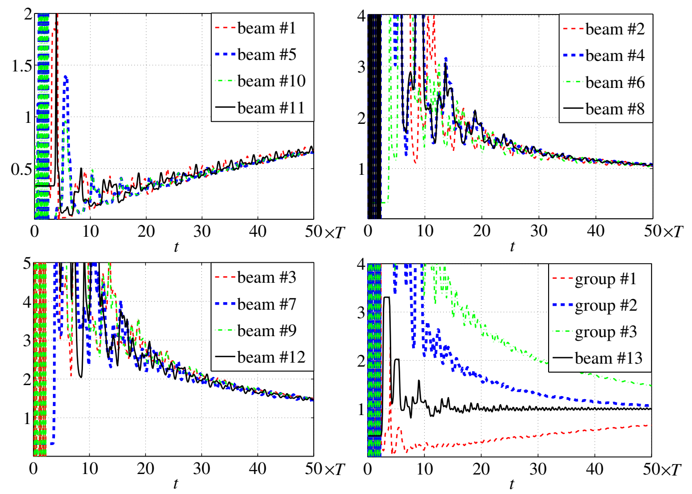

Figure 4. Evolution of the ratio $\dfrac{c_{\rm T}\mathscr{E}_{\rm T}^p(t)}{2c_{\rm P}\mathscr{E}_{\rm P}^p(t)}$ in each beam group.

from the multiple scattering of waves on random heterogeneities or inclusions considered elsewhere (see [3, 11, 13]), but it gives rise to the a similar qualitative and quantitative behavior at late times. Diffusion is characterized by an equilibration of the total energy in each subsystem (a beam of the truss) and for each propagation mode, a situation comparable to the assumption of modal equipartition invoked in the Statistical Energy Analysis (SEA) of structural-acoustic systems. However a rigorous mathematical model of the diffusive behavior induced by multiple scattering at the junctions and interfaces lacks at present. Other improvements of the proposed model could be the refinement of the kinematic assumptions in order to describe warped cross-sections, or the consideration of prestressing forces. These issues are the subject of ongoing researches.

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